

Previously, electronic circuit simulation mostly focused on the analysis of electrical signals like voltage or current waveforms for analogue engineers and binary bit patterns for digital engineers. However, IC density has grown to billions of transistors, and now verification of system performance and reliability requires analysis of both thermal and electrical conduction, involving modelling of materials that were previously ignored. In this interview, four industry leaders discuss whether multiphysics simulation really helps design engineers. Sudhir Sharma of ANSYS, Valerio Marra of COMSOL, Dr Detlef Schneider of Altair Engineering, and Kapil Gaitonde of SolidWorks spoke to EFY in this special interview

If you had to list three features that are most necessary in a multi-physics simulation product today, what would they be?

Scalability. To solve increasingly complex models efficiently, it is important to leverage the growing number of available hardware/computer resources. Scalability also includes efficient handling of a growing volume of data.

High quality. Only a simulation model that captures physics correctly and where users can rely on the results, ought to be used to make decisions.

Robustness. Multiphysics simulation can be accomplished by methods ranging from direct integration in one solver, through one-directional coupling, to full co-simulation.

Accuracy is of paramount importance when it comes to verifying and optimising a design. Two additional features that are also vital are the ability to couple any physics, and interoperability.

Designers need to be able to trust simulation results, mimic what happens in the real world, and use external data resources in the form of CAD files or material properties, for example.

Imagine the design of a mobile phone. We need to ensure that the phone is cool to touch and the electronic components perform at an optimal level. We also need to ensure that electromagnetic interference doesn't degrade the performance of electronic circuits and doesn't cause interference with other electronic equipment. Last of all, mechanical simulation is required to ensure that all sub-components will physically fit inside the physical body of the product and also retain structural integrity under various operating conditions.

1. It would include elements that support all the physics at the same time. I am talking about the mathematics involved.

2. At the back end, these complex elements work to carry out the multiphysics calculations, and simple options based on common terminology are made available to the designer.

3. Such elements involve a huge amount of calculations that would take up a lot of time. Hence, such products need to be very efficient in using the best available hardware and should give relevant and accurate results very fast.

What are the major industry challenges encountered in electronic product design that concern simulation users the most?

From a structural perspective, these are optimised packaging and weight, thermal analysis, robust design and cost. Simulation gives designers the option to replace many of the physical tests with virtual validation and prototyping, thereby reducing costs and the time of development, as well as increasing the robustness of design.

The need to be able to do more simulation iterations within as short a product design and development cycle as possible requires access to HPC (high performance computing) systems, which can allow multiple simulation runs to be carried out. The more the number of iterations, the more are the insights into design validation.

From our experience we see a dramatic increase in the density of components housed in an electronic device. The biggest challenges are, as mentioned previously, thermal management and also reliability and packaging. All aspects of product design involving structural mechanics, electromagnetics, heat transfer, fluid flow and chemical reactions present unique challenges. When a simulation can combine all of these phenomena together, and correctly represent their interactions, designers can be assured that their products will hold up once they are out there in the real world.

New products are driven by elements of innovation, changes in industry standards or in the competitive landscape. These elements are captured in a text document called 'Product specifications.' If the specification is misinterpreted by engineers, or continuously changed because of external factors, then projects are delayed, are of lower quality, have higher development costs, and may entirely miss the market. To prevent this, engineers need to adopt high-fidelity simulation tools that capture the design intent.

I think one of the primary challenges is the skillsets currently available. Most of the designers in this industry do not have a good understanding of finite element analysis (FEA), computational fluid dynamics (CFD), etc. However, they are very good in their domain and understand electronics very well. So, obviously, this requires tools that speak their language and do the FEA and CFD calculations at the back end.